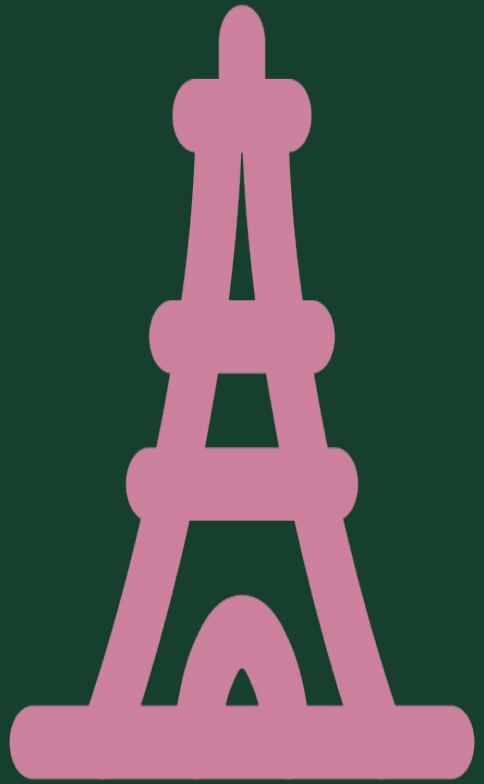
Bonjour, GrandPied !

SpamJam

Catching spam without losing a client email



Victoria Alajemba & Sheila Kana · ML × Python final project · Souheil Hanoune · November 2026

Le problème

Blocking a client email is the costly mistake



False positive

Costly

A client email lands in the spam folder: a missed inquiry, maybe a lost contract.



False negative

Acceptable

A spam reaches the inbox: annoying, but staff can delete it.

So we choose models by **average precision**, then set the decision threshold to keep precision on spam high.

Les données

Spambase: 4,210 unique emails, 57 numeric features

4,601

emails in Spambase (UCI, 1999)

391

exact duplicates removed

40 %

spam, kept in both sets by a stratified split

3,368 / 842

training / test emails (80/20, seed 42)

Watch out: all legitimate emails come from one person (George, at HP Labs). Words like george, hp or 650 describe him, not spam.

Also: capital-letter counts are very skewed (so we log them), and 857 and 415 are near-duplicates.

La méthode

One pipeline, tuned on the training set only



once, before training

inside the scikit-learn Pipeline, learned per CV fold

Validation

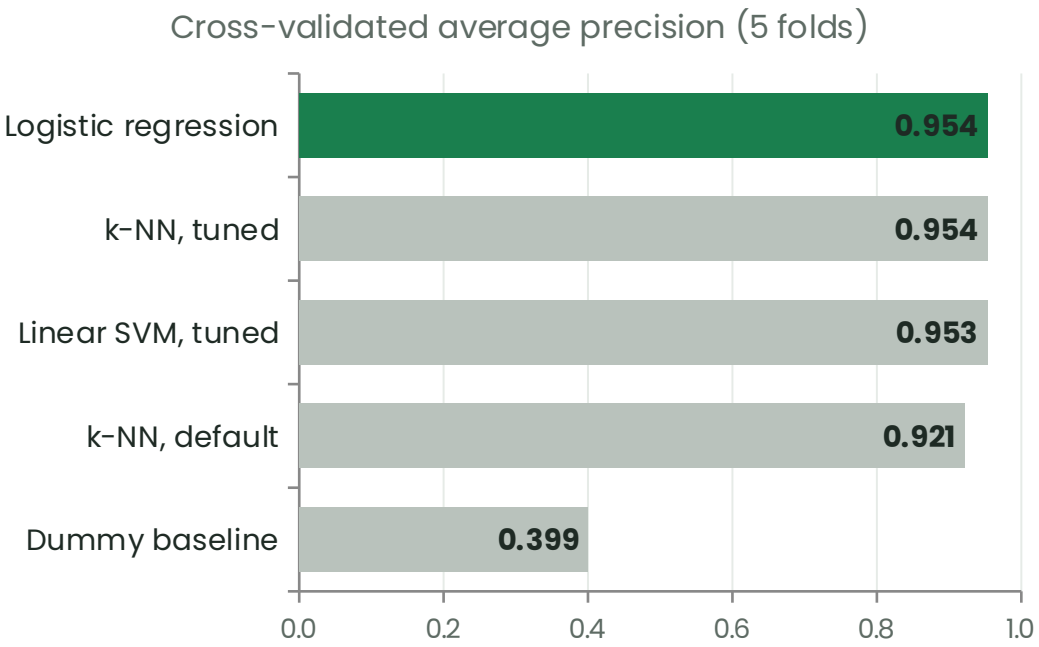
- Stratified 5-fold CV + GridSearchCV, scored by average precision
- A model wins only by > 1 standard error
- Threshold from out-of-fold predictions
- Test set used once, at the very end

Models compared

- Dummy baseline (always ham)
- Logistic regression (L1/L2, C)
- k-NN (neighbours, weighting)
- Linear SVM (C)

Les résultats

Three models tied: we keep the simplest



Why logistic regression

- All three within 0.001: far below one standard error (0.005)
- Simplest: probabilities and readable weights

Along the way

- Log on capitals: +0.011
- Tuning k-NN: 0.921 → 0.954

Le test final

On unseen emails, 95 % of flagged emails are spam

0.782
decision threshold, from out-of-fold predictions

0.950
precision: flagged emails that are spam

0.795
recall: about 4 spam out of 5 caught

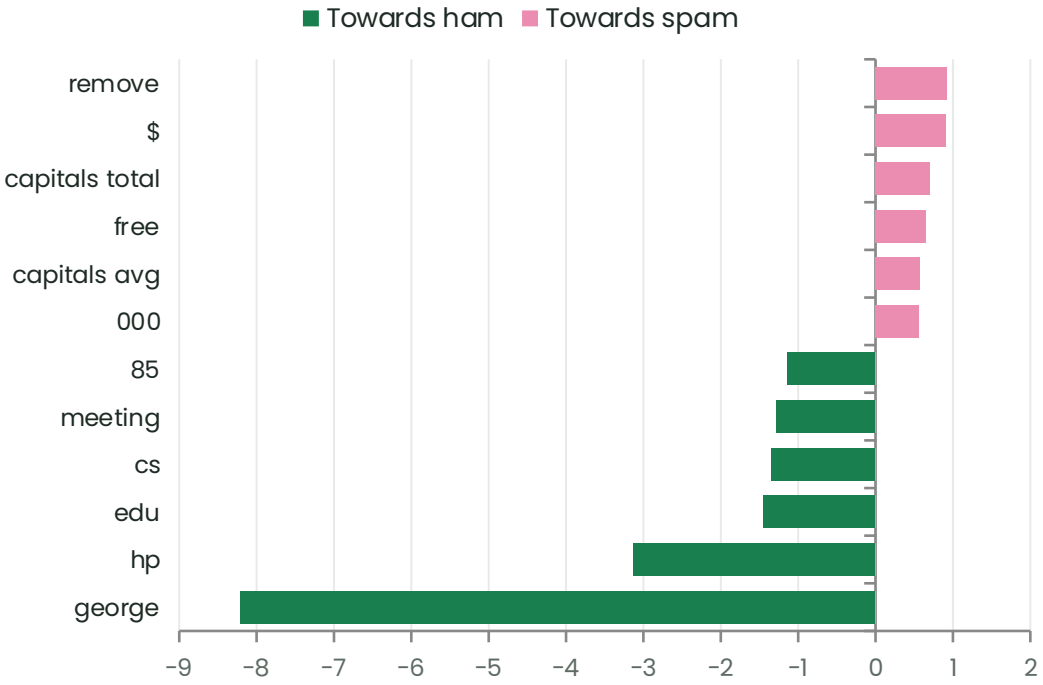
14 / 506
legitimate emails flagged

	predicted ham	predicted spam
actual ham	492	14
actual spam	69	267

The test set was used once, at the end. Precision matches the 0.95 we aimed for (margin 0.92 to 0.97). Without the donor's words: same precision (0.952), recall 0.71 instead of 0.79. Flagged emails go to quarantine, not the bin.

Ce que le modèle a appris

The donor's name is the model's strongest signal



The donor effect

Without the 9 donor words, average precision drops from 0.954 to 0.938 (about 3 standard errors).

That gain only exists in George's mailbox.

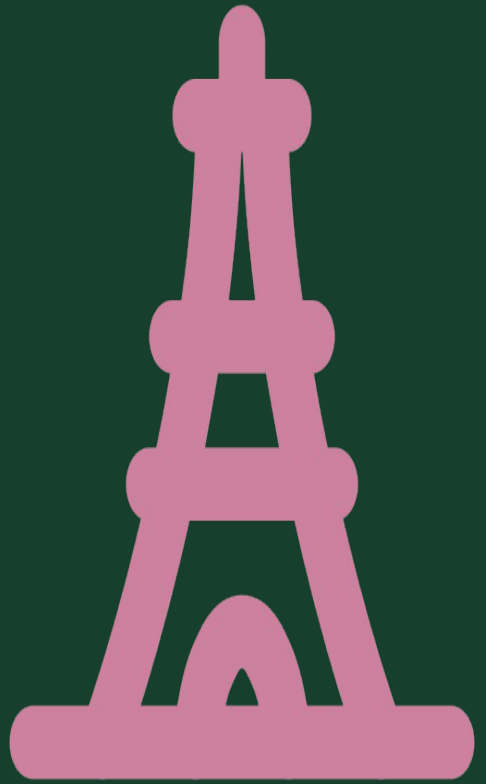
Trained without them, the model isn't fooled by "Hi George," (0 % → 100 % spam).

GrandPied must retrain on its own emails.

Merci !

To sum up

- Python package + CLI: train · evaluate · predict
- 67 tests, a Streamlit page, works on a fresh install
- Precision 0.95 on unseen emails, at threshold 0.782
- Limits: one donor's mailbox, English only
- **Before real use: retrain on GrandPied's own emails**



Questions ?